Frames and matrix designs

Sharp sampling complexity for Walsh restricted isometries

Difficulty: extreme **Importance:** broadly interesting

Status: Open

Rating rationale: Closing the uniform logarithmic sampling gap is a longstanding restricted-isometry barrier comparable in scale to the cyclic Fourier problem, with consequences for sparse recovery and fast sketches.

Let $N = 2^d$ with $d \geq 1$ and index rows and columns by $\mathbb{F}_2^d$. The normalized Walsh matrix is

$$(H_N)_{a,b} = N^{-1/2}(-1)^{a\cdot b}.$$

For $m \geq 1$, choose row indices $a_1, \ldots, a_m$ independently and uniformly with replacement and set $\Phi = \sqrt{N/m}(H_N)_{(a_1,\ldots,a_m),:}$. Let $E_{m,N,k}$ be the event that

$$\tfrac{1}{2}\|x\|_2^2 \leq \|\Phi x\|_2^2 \leq \tfrac{3}{2}\|x\|_2^2 \quad \text{for every } x \in \mathbb{R}^N \text{ with } |\operatorname{supp} x| \leq k.$$

Define

$$m_*(N,k) = \min\{m \in \mathbb{N} : m \geq 1,\ \Pr(E_{m,N,k}) \geq 0.9\}.$$

Problem. Determine $m_*(N,k)$ up to universal multiplicative constants, uniformly for $1 \leq k \leq N$, including all necessary logarithmic factors. The same sample must work for all sparse vectors. This is a quantitative restatement of the published sampling gap, not a separately numbered conjecture; the success probability and distortion fix a convention.

The Walsh transform is the Fourier transform on $\mathbb{F}_2^d$. Its subspace structure differs from the cyclic Fourier ensemble in FR-02, so their lower bounds cannot be interchanged.

References

1. J. Błasiok, P. Lopatto, K. Luh, J. Marcinek, and S. Rao, *An improved lower bound for sparse reconstruction from subsampled Walsh matrices*, Discrete Analysis 2023:3, introduction and Theorem 3.1. Paper.
2. I. Haviv and O. Regev, *The restricted isometry property of subsampled Fourier matrices*, Theorem 1.1 and its bounded-orthonormal-matrix scope. Paper.

Status check — 2026-09-10

The lower-bound paper uses independent Bernoulli row inclusion; the upper-bound literature also treats independent draws with replacement, the explicit convention here. Its $\Omega(k \log k \log(N/k))$ obstruction applies in a specified intermediate sparsity range, not as an all-parameter formula. The logarithmic gap discussed there remains unclosed in the targeted search for later Walsh RIP results. Endpoint regimes must also be accounted for; in particular $m_*(N,1) = 1$.

Audit update (2026-09-10): Rechecked the Walsh lower-bound record and Haviv–Regev upper bound, and searched for a later matching rate. The intermediate-sparsity obstruction is not an all-parameter formula. Difficulty is aligned with the comparable Fourier gap in FR-02. This is a bounded literature check, not a proof that no solution exists.